Senior Project

CryptoCredit Bank

A complete decentralized finance (DeFi) credit platform. It offers multi-asset collateral, flexible credit tiers, and an integrated marketplace.

By: Daksh Khurana

OVERVIEW

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Multi-Asset Collateral

Deposit ETH, WBTC, and USDX as collateral.

Integrated Marketplace

Shop using crypto payments and asset-backed financing.

Flexible Borrowing

Borrow USDX, ETH, or WBTC with deposited collateral.

Credit Tier System

Unlock better loan terms with consistent repayment.

TECH STACK

Our Tech Stack

Smart Contracts

Solidity 0.8.20, Hardhat 2.28, OpenZeppelin 5.4

Frontend

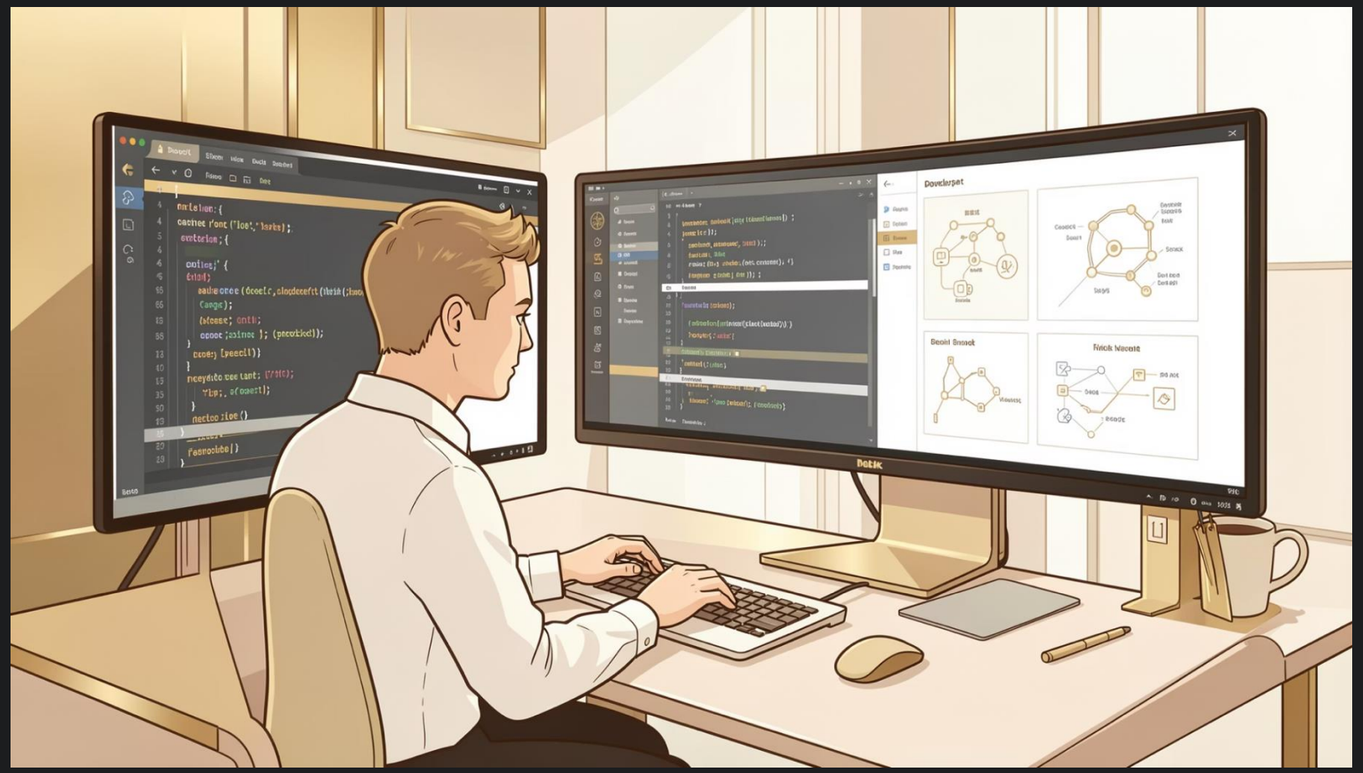
React 19, Vite 7, wagmi 2.19, RainbowKit 2.2, ethers.js 6

Backend

Express.js 5, MongoDB 7 (optional), WebSocket

DevOps

Docker, docker-compose



SETUP

Setup Essentials

Requirements

- **Local Development:** Node.js v18/v20, npm v9+, Git.
- **Docker:** Docker Desktop v4+ with Docker Compose v2 (no Node.js needed).
- **Wallet Connection (Optional):** MetaMask browser extension.

Installation

- **F ŒŐŌÑĪ ÑŐŐ**

  `git clone https://github.com/1n1t6Sh3ll/SeniorProject && cd SeniorProject/app`

- **Install**

  Main App: `npm install`

  Frontend: `cd web && npm install && cd ..`

- **Configure Environment**

  `cp .env.example .env` (for optional TestNet).

RUN OPTIONS

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Quick Dev Start

For fast development: `npm start` fires up Hardhat, deploys contracts, and launches the Vite dev server with hot-reload.

Docker for Sharing

Share with ease: `docker compose up --build` builds a single container with all services. No local Node.js needed.

Advanced Manual

For deep dives: Run blockchain, contract deployment, frontend, and backend services in separate terminals.

WALLET INTEGRATION

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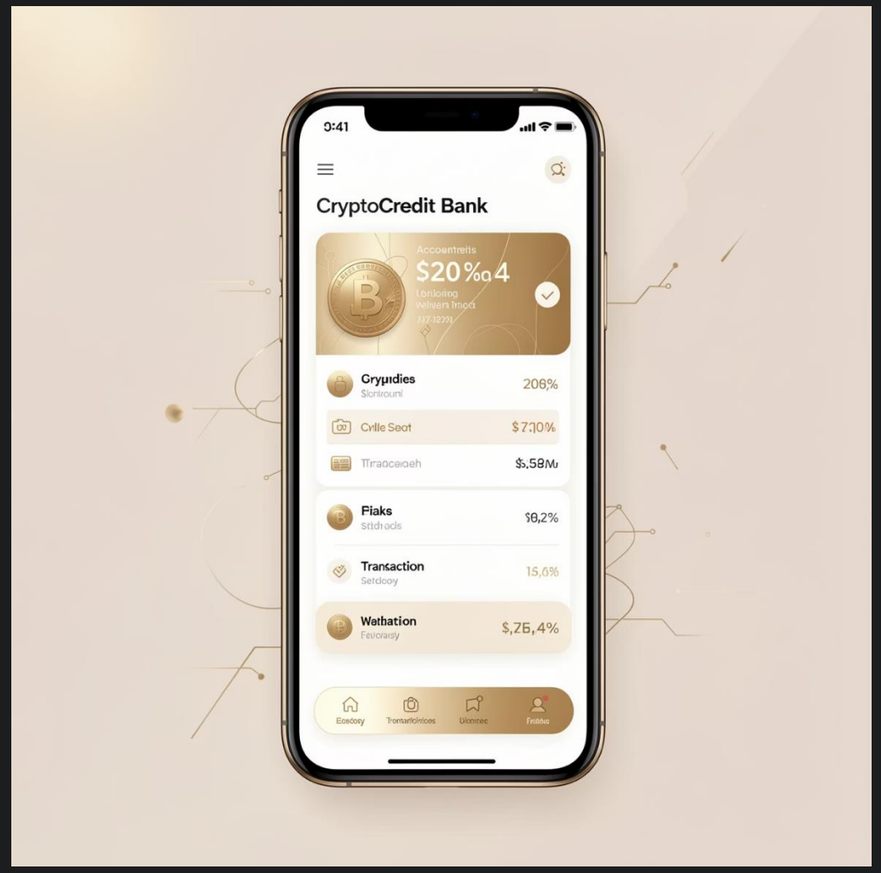
CryptoCredit Wallet

For a smooth experience, use our built-in **CryptoCredit Bank wallet**, designed for direct integration with our application.

MetaMask

MetaMask is fully supported for external wallet users. Connect with these details:

- RPC URL: `http://127.0.0.1:8545` or `http://localhost:8545`
- Chain ID: `31337`
- Import Private Key (Account #0): `0xac09...f2ff80`



CORE MECHANICS

Smart Contracts: The Protocol in Action

Five Solidity contracts manage all multi-asset lending and swapping.

CreditProtocol.sol

Handles lending logic for deposits, borrows, repayments, and liquidations.

USDX.sol

Our ERC-20 stablecoin, minted/burned by the protocol for borrowing/repayment.

MockWBTC.sol

Mock Wrapped Bitcoin, with a faucet for testing.

MockPriceOracle.sol

Delivers real-time ETH and WBTC price feeds (8 decimal precision).

SimpleSwap.sol

A Decentralized Exchange (DEX) for ETH, WBTC, and USDX token swaps.

USER BENEFITS

Unlock Better Loan Terms

Boost your credit with repayments for better loan terms.

Tier	Repay ments	Interest Rate	LTV Bonus
Bronze	0+	12% APR	+0%
Silver	3+	8% APR	+5%
Gold	7+	5% APR	+10%

Track your progress with the **CreditTierRoadmap**.

FEATURES

Your CryptoCredit Hub: Borrow, Shop, and Manage

CryptoCredit gives you essential tools to navigate DeFi.

Marketplace

Shop for assets with crypto or financing.

Exchange (DEX)

Trade ETH, WBTC, USDX at live rates.

Address Book & Limits

Manage contacts and set transfer limits.

Account Statements

View and export monthly statements.

Credit Score

Your credit score (300-850) shapes loan terms.

Interest & Fees

Understand daily interest, origination, and withdrawal fees.

SECURITY

Secure by Design

Our protocol protects your assets and data with strong security.

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Code Security

- **ReentrancyGuard:** Blocks common vulnerabilities.
- **SafeERC20:** Ensures safe token transfers.
- **Access Control:** Controls critical functions.
- **Health Factor Checks:** Prevents unsafe withdrawals and borrows.

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Heads Up

Never use real money in test environments, as keys are public.

CONCLUSION

CryptoCredit Bank: The Future of DeFi Credit

CryptoCredit Bank is a cool, multi-asset lending platform designed to empower users with flexible borrowing options and a transparent credit tier system. By leveraging cutting-edge smart contract technology, we provide a secure and integrated financial ecosystem.

> "We believe in a decentralized future where credit is accessible, fair, and user-centric."

Thank you for your time and interest in the CryptoCredit Bank Senior Project.

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